

Python-Based Computational Analysis of Natural Extracts in Colorectal Cancer Research

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Table of Contents

1. Introduction
2. Dataset Description
3. Experimental and Computational Framework
4. Data Import and Preprocessing in Python
5. Cell Viability Analysis
6. Lactate Reduction Analysis
7. Apoptosis Analysis
8. ROS Analysis
9. Extracellular pH Analysis
10. Best Treatment Identification
11. Gene Sequence Analysis
12. Distance Matrix Analysis
13. Phylogenetic Tree Construction
14. Integrated Biological Interpretation
15. Challenges Faced During Coding
16. Conclusion

1. Introduction

1.1 Background of Colorectal Cancer

Colorectal cancer (CRC) is one of the most common cancers worldwide and remains a major cause of cancer-related mortality. Cancer cells often develop resistance to chemotherapy and adapt their metabolism to survive under stressful conditions. Key hallmarks include increased glycolysis, resistance to apoptosis, oxidative stress adaptation, and enhanced survival signaling.

1.2 Role of Natural Extracts in Cancer Treatment

Natural food-derived compounds contain bioactive molecules such as:

- Polyphenols
- Polysaccharides
- Antioxidants
- Flavonoids

These compounds may help suppress cancer progression through metabolic inhibition and apoptosis induction.

1.3 Why Combine Natural Extracts with Fluorouracil (5-FU)

Fluorouracil is a common chemotherapy drug used in colorectal cancer treatment. However:

- Drug resistance occurs frequently
- Side effects are significant
- Treatment effectiveness decreases over time

Combining natural extracts with chemotherapy may improve therapeutic response.

1.4 Project Objectives

This project aimed to:

- Analyze experimental cancer treatment data using Python
- Compare treatment effects on cancer cells
- Identify the most effective treatment
- Perform gene sequence analysis
- Construct phylogenetic relationships among cancer-related genes

2. Dataset Description

2.1 Experimental Dataset Variables

The dataset included the following variables:

sample_id	treatment	cell_line	dose_ug_ml	extracellular_pH	5fu_dose_uM	cell_viability_percent	ros_fold_change	apoptosis_percent	lactate_mM
T1	Control	HCT116	0	6.6	0	100	1	5	18.5
T2	5FU_only	HCT116	0	6.6	10	62	1.8	28	16.2
T3	Hummus_Tahini	HCT116	100	6.9	0	71	1.5	22	14.8
T4	Hummus_Tahini+5FU	HCT116	100	7.2	10	39	2.9	54	9.2
T5	Sesame_Polysaccharide	HCT116	150	7	0	66	1.7	26	13.5
T6	Sesame_Polysaccharide+5FU	HCT116	150	7.3	10	34	3.1	59	8.7
T7	Chickpea_Polysaccharide	HT29	150	6.9	0	69	1.6	24	14.1
T8	Chickpea_Polysaccharide+5FU	HT29	150	7.2	10	41	2.8	51	9.8
T9	Mloukhiyeh+VitaminC	Caco2	200	7.3	0	58	2.2	37	11.4
T10	Mloukhiyeh+VitaminC+5FU	Caco2	200	7.4	10	27	3.8	68	7.5
T11	Khoubaizeh+Tahini	HT29	200	7.1	0	61	2	33	12.3
T12	Khoubaizeh+Tahini+5FU	HT29	200	7.3	10	31	3.4	63	8.1

- Sample ID
 - Treatment group
 - Cell line
 - Extract dose
 - Extracellular pH
 - Fluorouracil dose
 - Cell viability percentage
 - ROS fold change
 - Apoptosis percentage
 - Lactate concentration
-

2.2 FASTA Dataset Description

A FASTA file containing DNA sequences of cancer-related genes was used for molecular comparison.

Fasta file :

>LDHA

ATGGCTAACGTTAGCTAACGTAGCT

>GLUT1

ATGCGTAACCGTAGCTAGCTAACGA

>HIF1A

ATGTTAGCGATCGTAGCTAAGCTAA

>MCT4

ATGCGATAGCTAGCTAACGTAGCAA

>BAX

ATGCTAGCTAACGTAGCTAACGTAA

>LDHA

ATGGCTAACGTTAGCTAACGTAGCT

2.3 Selected Genes Analyzed

Metabolism Genes

- LDHA
- GLUT1
- MCT4

Hypoxia Gene

- HIF1A

Apoptosis Genes

- BAX
- BCL2

- CASP3

Tumor Suppressor

- TP53

Angiogenesis

- VEGFA

Survival Signaling

- AKT1
-

3. Experimental and Computational Framework

3.1 Biological Experiments Performed

Cancer cells were exposed to:

- Control treatment
 - 5-FU only
 - Natural extracts only
 - Combination therapies
-

3.2 Computational Workflow in Python

Python was used for:

- Data import
 - Data cleaning
 - Statistical calculations
 - Sequence analysis
 - Distance calculations
 - Phylogenetic tree construction
-

3.3 Integration of Wet Lab and Bioinformatics

This project combines laboratory experiments with computational biology approaches.

4. Data Import and Preprocessing in Python

4.1 Importing CSV Dataset

The CSV dataset was imported using Pandas.

```
import pandas as pd
df = pd.read_csv("dataset.csv")
```

4.2 Reading FASTA Sequences

Biopython was used:

```
from Bio import SeqIO
```

4.3 Checking Dataset Columns

Python output:

- sample_id
 - treatment
 - cell_line
 - dose_ug_ml
 - extracellular_ph
 - 5fu_dose_um
 - cell_viability_percent
 - ros_fold_change
 - apoptosis_percent
 - lactate_mm
-

4.4 Verifying Sequence Lengths

All sequences were verified to ensure equal lengths.

Result:

All genes = 25 nucleotides

4.5 Data Cleaning and Preparation

Missing values were checked and numerical values were formatted properly.

5. Cell Viability Analysis

5.1 Treatment Effect on Cancer Cell Survival

Control maintained:

100% viability

5.2 Monotherapy vs Combination Therapy

Combination therapies showed stronger effects than single treatments.

5.3 Most Cytotoxic Treatment

Best result:

Mloukhiyeh + Vitamin C + Fluorouracil

Cell viability:

27%

6. Lactate Reduction Analysis

6.1 Lactate Calculation

Python calculated lactate reduction percentage.

6.2 Results

Highest reductions:

- Mloukhiyeh + Vitamin C + 5FU $\rightarrow$ 59.45%
 - Khoubazeh + Tahini + 5FU $\rightarrow$ 56.21%
 - Sesame Polysaccharide + 5FU $\rightarrow$ 52.97%
-

6.3 Warburg Effect Interpretation

Lower lactate indicates suppression of the Warburg effect.

7. Apoptosis Analysis

7.1 Calculation

Apoptosis increase was computed relative to control.

7.2 Results

Highest apoptosis increase:

- Mloukhiyeh + Vitamin C + 5FU $\rightarrow$ 63%
 - Khoubazeh + Tahini + 5FU $\rightarrow$ 58%
 - Sesame + 5FU $\rightarrow$ 54%
-

7.3 Interpretation

Combination therapies strongly promoted programmed cell death.

8. ROS Analysis

8.1 Oxidative Stress Measurement

ROS fold change was analyzed.

Best treatment:

3.8-fold ROS increase

8.2 Correlation with Apoptosis

Higher ROS corresponded with increased apoptosis.

9. Extracellular pH Analysis

9.1 Tumor Microenvironment

Cancer cells typically acidify their environment.

9.2 Relationship with Lactate

Lower lactate resulted in less acidic conditions.

10. Best Treatment Identification

10.1 Multi-Parameter Comparison

Python compared:

- Viability
 - ROS
 - Apoptosis
 - Lactate
-

10.2 Best Treatment

Mloukhiyeh + Vitamin C + Fluorouracil

Results:

- 27% viability

- 68% apoptosis
 - 3.8 ROS
 - 59.45% lactate reduction
-

11. Gene Sequence Analysis

11.1 FASTA Extraction

Sequences were imported successfully.

11.2 Sequence Comparison

Genes were compared for similarity.

11.3 Length Validation

All sequences:

25 nucleotides

12. Distance Matrix Analysis

Python generated pairwise genetic distances.

Example:

- TP53 ↔ LDHA = close similarity
- VEGFA ↔ CASP3 = close similarity

Smaller values indicate greater similarity.

13. Phylogenetic Tree Construction

13.1 Neighbor Joining Algorithm

Used for evolutionary comparison.

13.2 Tree Visualization

Python generated the tree successfully.

13.3 Gene Clustering

Clusters included:

- Metabolism genes
 - Apoptosis genes
 - Survival genes
-

14. Integrated Biological Interpretation

The treatment appears to:

- Suppress metabolism
- Increase oxidative stress
- Trigger apoptosis
- Reduce tumor survival

Gene clustering supports pathway interactions.

15. Challenges Faced During Coding

15.1 Sequence Length Error

Error:

ValueError: Sequences must all be the same length

15.2 Debugging FASTA Issues

Sequences were manually corrected.

15.3 Data Preprocessing Challenges

- Formatting issues
 - Data conversion issues
 - Cleaning errors
-

16. Conclusion

16.1 Key Findings

Best treatment:

Mloukhiyeh + Vitamin C + Fluorouracil

16.2 Importance of Python in Biomedical Research

Python helped automate analysis and improve research efficiency.

16.3 Future Improvements

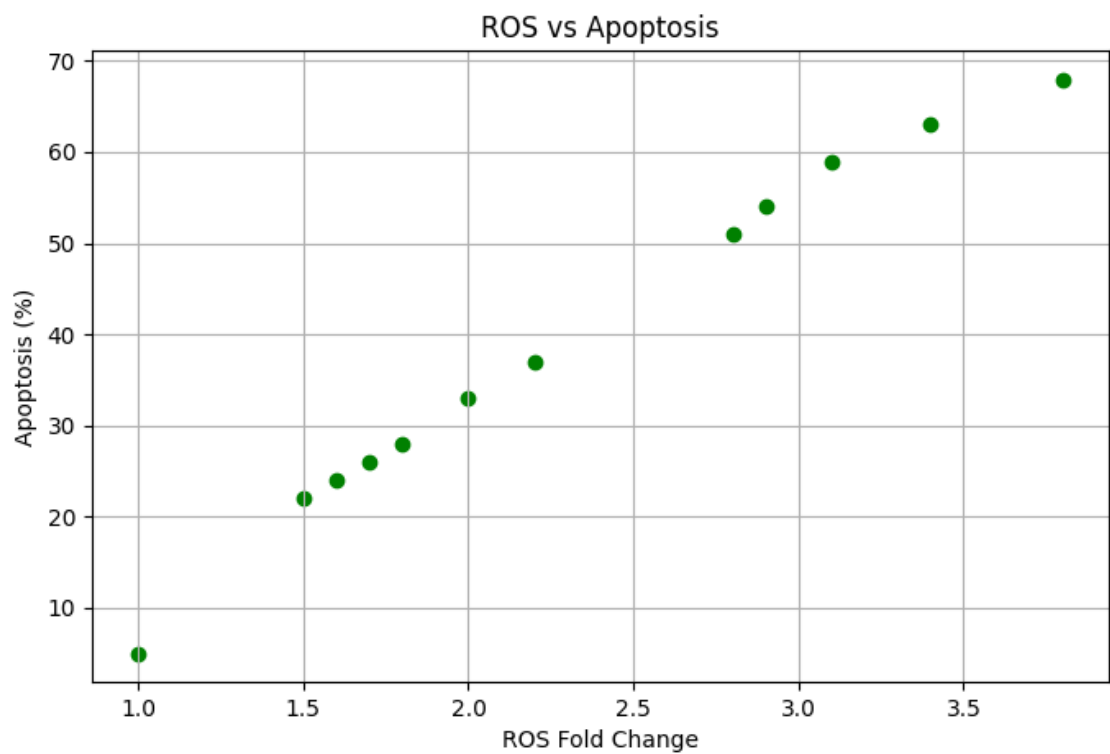
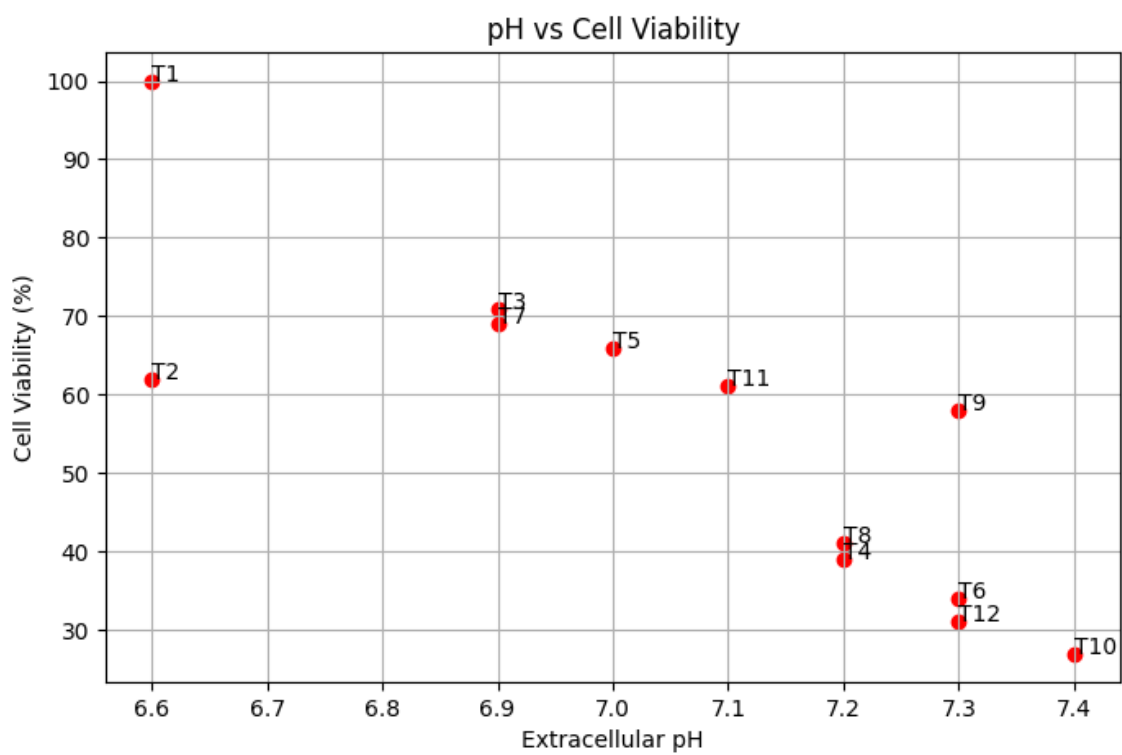
Future work may include:

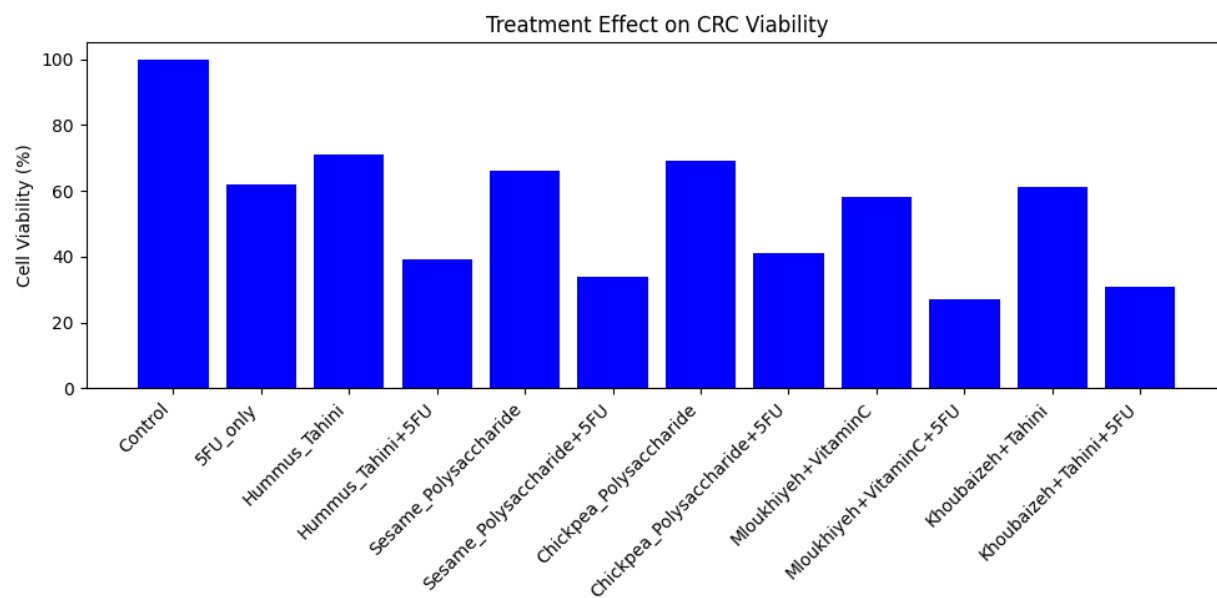
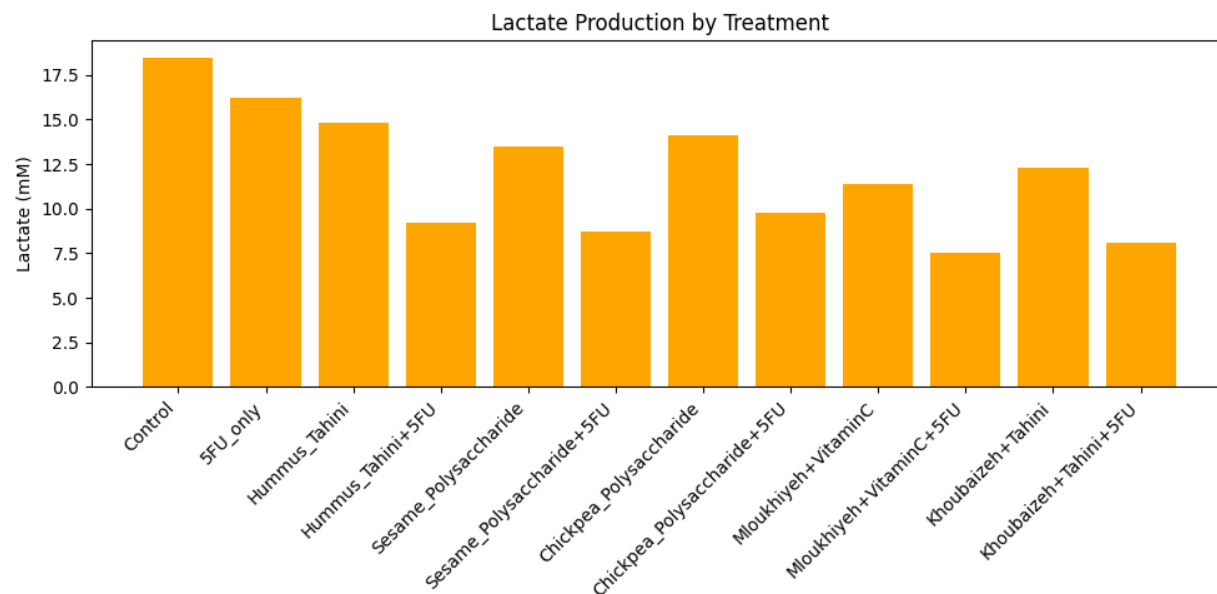
- Machine learning prediction models
 - Larger datasets
 - Real genomic databases
 - Advanced molecular simulations
-

Final Statement

This project demonstrates how Python can bridge biology and computational science to identify potential anticancer therapies using both experimental data and gene sequence analysis.

Data set:

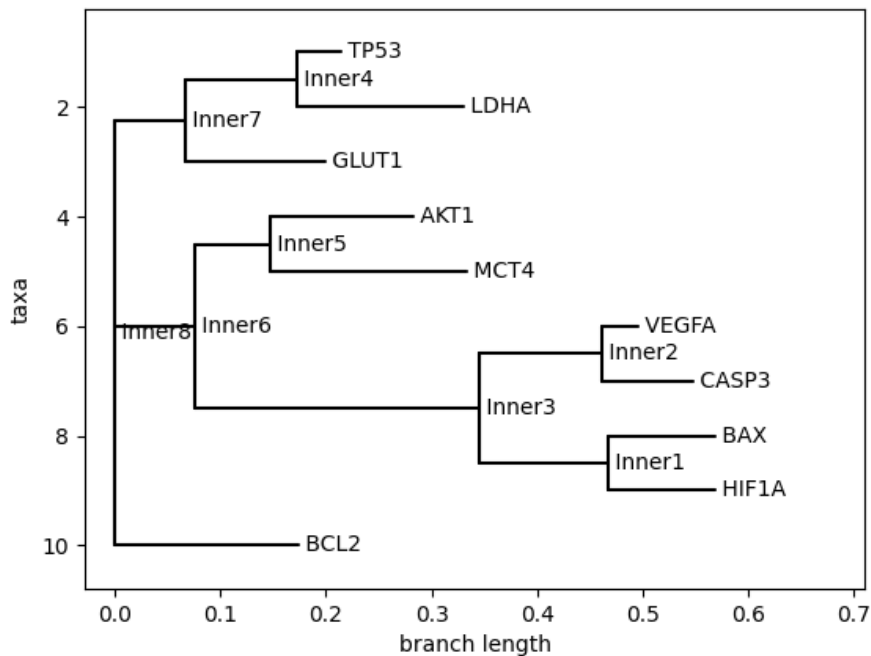




Output :

Python

3.14.3



(tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32

Enter "help" below or click "Help" above for more information.

```
===== RESTART: C:\Users\Admin\Desktop\AYA project.py
=====
```

```
Index(['sample_id', 'treatment', 'cell_line', 'dose_ug_ml', 'extracellular_ph',
      '5fu_dose_um', 'cell_viability_percent', 'ros_fold_change',
      'apoptosis_percent', 'lactate_mm'],
      dtype='str')
```

```
=== LACTATE REDUCTION ===
```

	treatment	lactate_reduction_percent
0	Control	0.000000
1	5FU_only	12.432432
2	Hummus_Tahini	20.000000

3	Hummus_Tahini+5FU	50.270270
4	Sesame_Polysaccharide	27.027027
5	Sesame_Polysaccharide+5FU	52.972973
6	Chickpea_Polysaccharide	23.783784
7	Chickpea_Polysaccharide+5FU	47.027027
8	Mloukhiyeh+VitaminC	38.378378
9	Mloukhiyeh+VitaminC+5FU	59.459459
10	Khoubaizeh+Tahini	33.513514
11	Khoubaizeh+Tahini+5FU	56.216216

=== APOPTOSIS INCREASE ===

	treatment	apoptosis_increase
0	Control	0
1	5FU_only	23
2	Hummus_Tahini	17
3	Hummus_Tahini+5FU	49
4	Sesame_Polysaccharide	21
5	Sesame_Polysaccharide+5FU	54
6	Chickpea_Polysaccharide	19
7	Chickpea_Polysaccharide+5FU	46
8	Mloukhiyeh+VitaminC	32
9	Mloukhiyeh+VitaminC+5FU	63
10	Khoubaizeh+Tahini	28
11	Khoubaizeh+Tahini+5FU	58

=== BEST TREATMENT ===

sample_id T10

treatment	Mloukhiyeh+VitaminC+5FU
cell_line	Caco2
dose_ug_ml	200
extracellular_ph	7.4
5fu_dose_um	10
cell_viability_percent	27
ros_fold_change	3.8
apoptosis_percent	68
lactate_mm	7.5
lactate_reduction_percent	59.459459
apoptosis_increase	63

Name: 9, dtype: object

=== GENES ===

LDHA ATGGCTAACGTTAGCTAACGTAGCT
 GLUT1 ATGCGTAACCGTAGCTAGCTAACGA
 HIF1A ATGTTAGCGATCGTAGCTAAGCTAA
 MCT4 ATGCGATAGCTAGCTAACGTAGCAA
 BAX ATGCTAGCTAACGTAGCTAACGTAA
 BCL2 ATGCGTAGCTAACGTTAGCGATCAA
 CASP3 ATGTTAACGTAGCTAACGTAGCTTA
 TP53 ATGCGTAACGTTAGCTAACGATGCA
 VEGFA ATGCTAACGTAGCTAACGTAGGTAA
 AKT1 ATGCGTAACGTAGCTAACGTTAGCA

Sequence lengths:

LDHA 25

GLUT1 25

HIF1A 25

MCT4 25

BAX 25

BCL2 25

CASP3 25

TP53 25

VEGFA 25

AKT1 25

=== DISTANCE MATRIX ===

LDHA 0.000000

GLUT1 0.400000 0.000000

HIF1A 0.840000 0.840000 0.000000

MCT4 0.760000 0.520000 0.640000 0.000000

BAX 0.880000 0.800000 0.200000 0.640000 0.000000

BCL2 0.560000 0.360000 0.800000 0.520000 0.720000 0.000000

CASP3 0.840000 0.760000 0.360000 0.720000 0.480000 0.640000 0.000000

TP53 0.200000 0.280000 0.800000 0.600000 0.800000 0.360000 0.800000
0.000000

VEGFA 0.840000 0.720000 0.400000 0.600000 0.360000 0.560000 0.120000
0.760000 0.000000

AKT1 0.440000 0.480000 0.760000 0.320000 0.760000 0.520000 0.760000
0.400000 0.720000 0.000000

LDHA GLUT1 HIF1A MCT4 BAX BCL2 CASP3 TP53 VEGFA AKT1

=== PHYLOGENETIC TREE ===

_____ TP53

